

Gene Pulser® Electroprotocols

* We recommend adapting this protocol to use the Gene Pulser electroporation buffer (catalog #165-2676, 165-2677), which increases cell viability and transfection efficiency in mammalian cell lines.

Cell Type	Mammalian, adherent, suspension	Molecules Electroporated	DNA, p220LTR, 11.2 kb, supercoiled.
Species Used	Human, 293s, transformed primary embryonic kidney; Lymphoblast cell lines, EBV immortalized		

Before the Pulse

Cell Growth Medium	RPMI, 10% Fetal Calf Serum, Penicillin, Streptomycin, L-Glutamine, or DMEM (GIBCO/BRL, Sigma)	Growth Phase at Harvest	80% confluent or 1 x 10 ⁶ cells / ml
Wash Solution	Electroporation Buffer (see notes)	Pre-pulse Incubation	None

The Pulse

Instruments Used Gene Pulser® apparatus & Capacitance

Electroporation Temperature	25 °C	Cuvette Gap	0.4 cm
Electroporation Medium*	See notes	Voltage	0.260 kV
Cell Density	1 x 10 ⁷ cells / µl	Field Strength	0.65 kV/cm
Volume of Cells	800 µl	Capacitor	550 or 960 µF
DNA Concentration	1 to 4 µg / µl	Resistor	(Pulse Controller) Ω none
DNA Resuspension Buffer	TE Buffer (10 mM Tris, 1 mM EDTA, pH 8.0)	Time Constant	6.4 or 11.2 msec
Volume of DNA	<20 µl		

After the Pulse

Outgrowth Medium RPMI, 10% Fetal Calf Serum, Penicillin, Streptomycin, L-Glutamine, or DMEM

Relevant Publications and/or Comments

Note: exponential values designated in parentheses.

Outgrowth Temperature	37°C	* Electroporation buffer from Chu, <i>et. al.</i> , <i>NAR</i> 15 (3): 1311 (1987); 1x HeBS, 20 mM HEPES, pH 7.05, 137 mM NaCl, 5 mM KCl, 0.7 mM Na ₂ HPO ₄ , 6 mM dextrose.
Length of Incubation	Infinite	
Selection Method or Assay Used	Hygromycin resistance	
Electroporation Efficiency	4000 transformants / µg DNA	
Per Cent Survival	10%	

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Survey Number

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